Internalizing Negation-Gated Logical Rules into LLMs for Document-Level Relation Extraction

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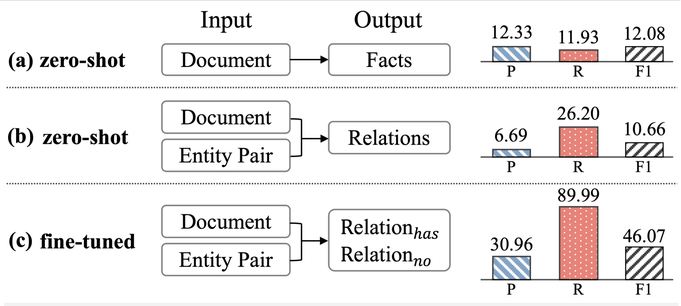
Motivation

[L1] Irrecoverable Pruning

Setting	DocRED dev		
	P	R	F1
Candidate Filter	44.82	66.56	53.57
Relation Extractor	89.93	89.52	89.72
Cascaded	41.27	61.68	49.45

89.72 F1 → 49.45 F1
in the cascaded pipeline
Hard candidate filtering permanently discards entity pairs predicted as NA, including implicit or intermediate facts needed for multi-hop reasoning

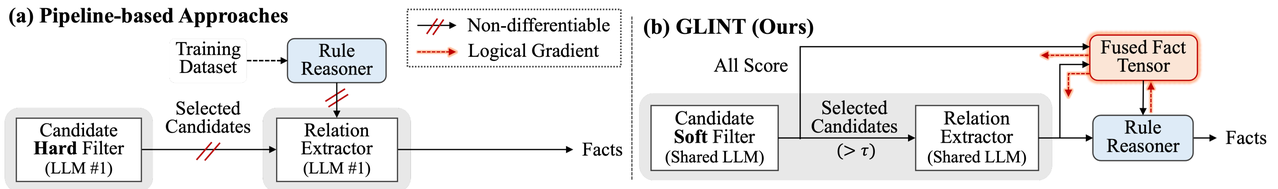
[L2] Spurious Over-Prediction



Pair-level querying → ↑ Recall, ↓ Precision
LLM priors induce unsupported predictions, yet existing methods lack explicit logical constraints to suppress them

Key Idea

: Internalize Logical Rules into the LLM itself

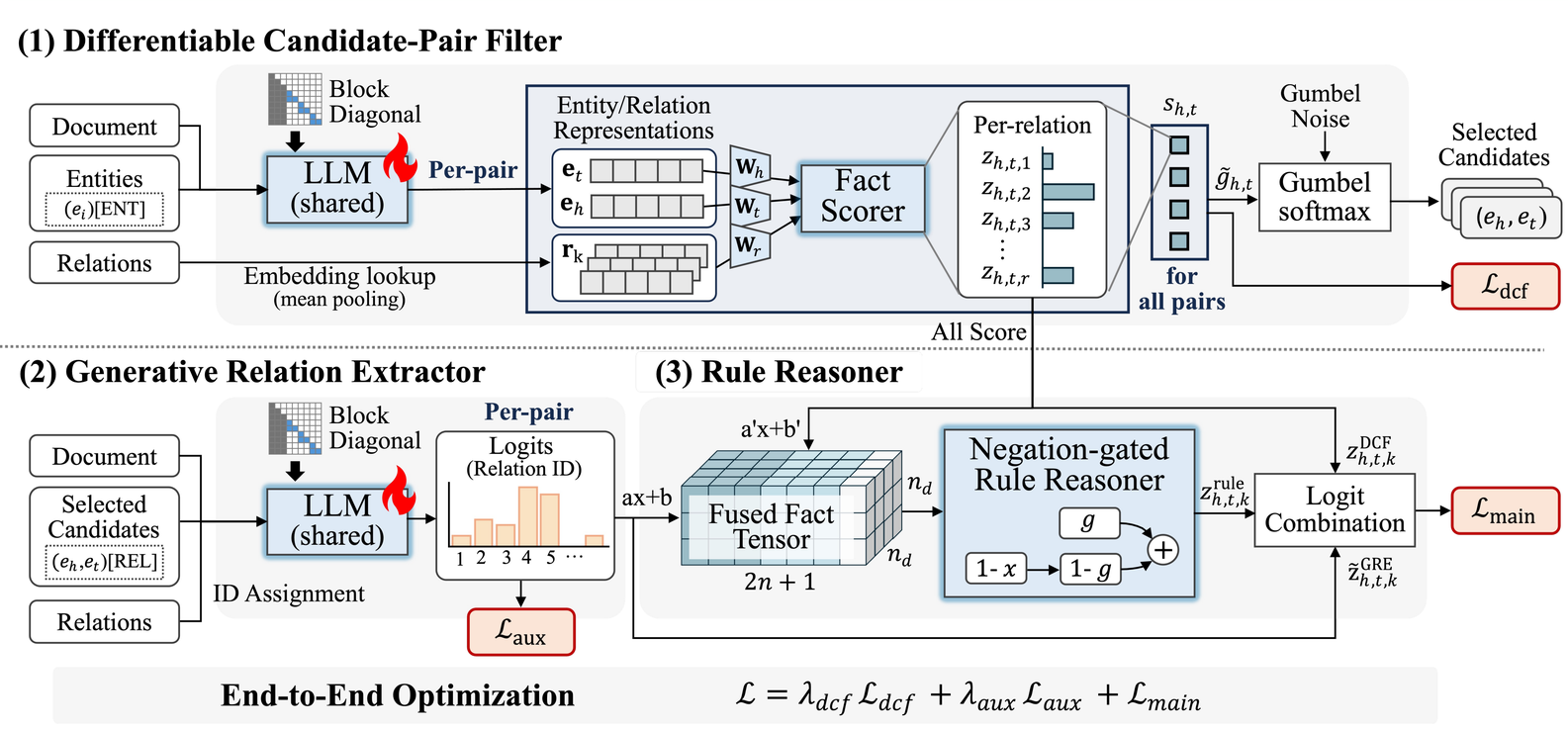


Challenges

[C1] Differentiable End-to-End Pathway
Preserve differentiability across candidate selection without $O(N^2)$ LLM queries

[C2] Unified Negation-aware Rule Learning
Incorporate positive and negative conditions without conflicting learning signals

Proposed Method: GLINT



Contributions

- **End-to-End LLM-Rule Integration**
: Jointly optimize the LLM backbone with inductive logical rule learning to propagate rule-augmented supervision
→ Experiment #1, #3
- **Differentiable Preservation of All Entity-Pair Signals**
: Replace irreversible hard pruning with differentiable selection while retaining signals from every entity pair
→ Experiment #2
- **Per-atom Learnable Negation Gates**
: Extend positive-only soft rules with learnable atom-level polarity to model positive, negative, and mixed logical conditions
→ Experiment #4

① Differentiable Candidate-Pair Filter

Goal: Identify promising entity pairs while prioritizing recall
- Compute per-relation scores for all entity pairs
- Aggregate them into pair-level scores and apply Gumbel-Softmax selection

=> Non-selected pairs retain dense DCF relation signals

② Generative Relation Extractor

Goal: Predict fine-grained relations for selected candidates
- Predict reserved relation tokens for each selected pair
- Pack multiple pair queries using block-diagonal attention

=> Produced sharp per-relation predictions while efficiently reusing the shared document prefix

③ Rule Reasoner over F+

Goal: Recover implicit relations and suppress unsupported predictions
- Fuse GRE scores with DCF non-candidate signals
- Learn positive or negated literals via per-atom gates

=> positive, negative, and mixed rules refine relation logits

Experiments

#1 Main Results

Model	DocRED				Re-DocRED			
	Dev		Test		Dev		Test	
	F1	Ign F1	F1	Ign F1	F1	Ign F1	F1	Ign F1
PromptRE	-	-	-	-	10.55	9.03	-	-
ChatGPT	32.21	-	-	-	28.89	-	-	-
DocGNRE	13.84	13.65	13.93	13.67	11.18	11.10	11.12	11.04
D-F	46.38	44.77	47.08	45.30	54.22	53.48	53.33	52.50
D-R-F	45.77	44.32	47.50	45.98	56.58	56.10	54.84	54.35
AutoRE	47.17	45.58	47.15	45.45	60.17	59.25	59.29	58.33
EP-RSR	53.77	51.25	54.57	51.77	65.14	63.93	64.24	63.03
LMRC	61.01	59.13	60.84	58.77	-	-	72.80	72.21
Ours (LLaMA3-8B)	61.75	59.42	62.38	59.86	77.47	76.32	77.33	76.19
Ours (LLaMA3.1-8B)	61.30	59.03	62.29	59.88	76.63	75.34	76.28	74.97

- GLINT achieves the best performance on both DocRED and Re-DocRED, improving Re-DocRED test F1 by +4.53 over the strongest LLM baseline
- Comparable gains across LLaMA3 and LLaMA3.1 indicate that the improvement comes primarily from the GLINT framework rather than a specific backbone

#2 Ablation Study

Model	Re-DocRED Dev		
	F1	Ign F1	ΔF1
Ours (GLINT)	77.47	76.32	-
w/o rule reasoner	73.86	72.54	-3.61
w/o negation	76.88	75.61	-0.59
w/o st-gumbel	76.16	75.93	-1.31
w/o phi	73.02	72.05	-4.45

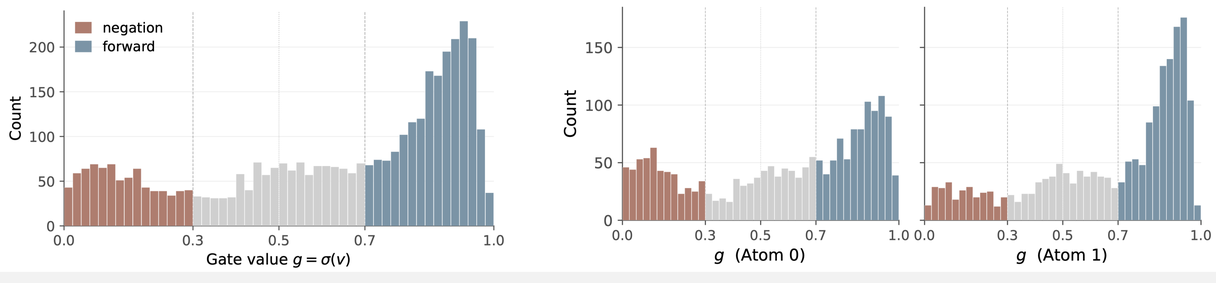
Preserving non-candidate signals (−4.45 F1) and rule reasoning (−3.61 F1) are the two largest contributors to GLINT

#3 Internalization Analysis

Setting	Re-DocRED				
	Dev		Test		
	F1	Ign F1	F1	Ign F1	ΔF1
Ours(GLINT)	77.47	76.32	77.33	76.19	+5.08
GLINT$_{zero}$	74.32	73.96	75.85	75.62	+3.60
GLINT$_{norule}$	73.86	72.54	72.25	71.93	-

GLINT$_{zero}$ retains +3.60 F1 over GLINT$_{norule}$, even with rule scores removed at inference
→ Rule-augmented training internalizes reasoning patterns into the LLM backbone

#4 Negation Gate Analysis



- 71.3% of gates move into clear forward or negated regimes rather than remaining neutral
- Negation is position-specific: 25.8% at Atom 0 vs. 14.4% at Atom 1